

Quantum Computing Foundations & Cybersecurity Implications

1. Quantum Computer Architecture & Core Principles

Quantum computers utilize qubits based on principles of quantum mechanics, rather than the binary bit system (0 and 1) used by classical computers.

Thanks to the principle of superposition, qubits can exist in states representing 0 and 1 simultaneously. Furthermore, quantum entanglement allows interconnected qubits to offer exponential parallel processing capability.

This technology possesses the potential to solve complex simulations and mathematical problems in minutes that would take classical supercomputers billions of years to compute.

2. Quantum Algorithms and Performance Analysis

Shor's Algorithm: Developed by Peter Shor in 1994, this algorithm solves the prime factorization problem at exponential speeds. Factoring an RSA-2048 key, which would take classical computers approximately 10,000 years, takes only 4 hours on a quantum computer with sufficient qubits.

Grover's Algorithm: Providing quadratic speedup for searching unsorted databases, Grover's Algorithm reduces effective key lengths of symmetric encryption algorithms like AES-256 down to 128 bits.

As a consequence, public-key cryptography standards currently securing the internet, such as RSA and ECC (Elliptic Curve Cryptography), will become entirely vulnerable to quantum adversaries.

3. Post-Quantum Cryptography (PQC)

To mitigate the collapse of public-key infrastructures, the National Institute of Standards and Technology (NIST) released official Post-Quantum Cryptography (PQC) standards.

PQC algorithms rely on hard lattice-based mathematical problems that are resistant to attacks from both classical and quantum computers. Prominent standards include CRYSTALS-Kyber (for key encapsulation) and CRYSTALS-Dilithium (for digital signatures).

Modernizing enterprise cybersecurity infrastructures to comply with these algorithms is referred to as achieving 'Quantum Resilience'.

4. Transition Timeline & 'Harvest Now, Decrypt Later' Threat

Cyber adversaries today execute 'Harvest Now, Decrypt Later' tactics. Attackers intercept and store encrypted sensitive corporate data today, intending to decrypt it once fault-tolerant quantum computers become available.

Consequently, the year 2030 has been established as the mandatory migration deadline for global organizations. Organizations are required to enforce AES-256 symmetric encryption and upgrade digital signatures to PQC standards.